Hariton **Pitsik**

Junior ML Engineer · ML Intern
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About Me

Fourth-year computer science student passionate about machine learning, with a focus on NLP, LLMs, and information retrieval. Build Python/PyTorch pipelines, evaluate model quality, and teach ML in a student community. Open to ML engineering roles and internships.

Skills

Programming languages	Python, SQL, Rust, Go
ML / DL	PyTorch, Scikit-learn, XGBoost, CatBoost, NumPy, Pandas, Polars
NLP / LLM	Transformers, RAG, Qdrant, FAISS, Ollama
Agentic systems	LangChain, LangGraph, LangSmith, MCP
Evaluation & experiments	Jupyter notebook, ranx, Optuna, ClearML, MLFlow
Engineering	Git & GitHub, Docker, Linux, FastAPI, Pydantic, PostgreSQL, Streamlit
Testing & CI	pytest, Unit/integration testing, GitHub Actions
Foundations	Linear algebra, Calculus, Probability & statistics, Supervised & unsupervised learning, NLP
Spoken languages	Russian (native), English (B2)

Practical Experience

Speech Recognition and Speaker Diarization

haritonn/asr_practice

University practical placement project.

- Built a Russian dialogue transcription pipeline with Faster-Whisper, Silero VAD, and pyannote; linked transcripts to speakers.
- Added domain-specific catalog-term recognition using a NeMo CTC context graph without ASR fine-tuning; evaluated WER, CER, DER, and terminology F1.

Projects

Local RAG for Scientific Papers

haritonn/qasper_rag_system

- Built document question answering on QASPER: paragraph indexing, BGE/BM42 embeddings via FastEmbed, Qdrant hybrid retrieval with RRF, cross-encoder reranking, and local generation through Ollama/Qwen3 8B.
- Evaluated within-paper retrieval on 888 of 1,005 validation questions: reranking raised MRR@10 from 0.4313 to 0.5221 and Recall@10 from 0.6932 to 0.7493.
- Separated retrieval and generation components behind interfaces; added source metadata in the CLI, unit/integration tests, GitHub Actions CI, and locked dependencies with uv.

Biomedical Search

haritonn/fitness_se

- Built search over 23,283 Europe PMC papers on training, recovery, and sports nutrition: weighted BM25, MedCPT/FAISS semantic search, and hybrid retrieval with RRF.
- Evaluated on 150 queries and 3,000 graded relevance judgments; BM25 reached test nDCG@10 0.5019. Added Ollama/Qwen3 8B-assisted relevance labeling, a Streamlit UI, and GitHub Actions CI.

MAS Crawler (WIP)

haritonn/mas_crawler

- Built an MVP for site-specific search: an LLM agent prepares a crawl plan from a URL and a user-defined goal; a Rust engine collects documents and builds a Tantivy/BM25 index. Integrated Python and Rust through PyO3 and orchestrated the pipeline with LangGraph.
- Implemented search-configuration comparison using nDCG, Precision, and Recall; the next stage is an agent that runs experiments and selects settings based on relevance judgments.

Image Captioning

haritonn/caption_gen

- Built a Flickr8k image-captioning pipeline in PyTorch: ResNet-50 encoder and soft-attention LSTM decoder; used scheduled sampling, label smoothing, and early stopping.
- Made experiments reproducible with deterministic splits, checkpoints, and optional ClearML tracking; achieved validation BLEU 0.1655 and METEOR 0.3774.

cargo-smi - NVIDIA GPU Monitor

haritonn/cargo-smi

- Built a Rust TUI for GPU and host monitoring using NVML, Ratatui, Crossterm, and Sysinfo: VRAM, utilization, temperature, CPU/RAM, and processes.
- Added multi-GPU navigation, configurable refresh, and utilization history charts.

Coursework

Bachelor's Thesis: GraphRAG for Question-Answering Systems Based on Documents and a Knowledge Graph.

WIP

3rd Year: Restoring the event hierarchy by clustering text embeddings

haritonn/coursework3

Investigated embedding geometry in news-event clustering: 2,604 HDBSCAN experiments across 3 embedding families, 93 feature spaces, and 28 parameter values. UMAP improved the best F1 for every family; Jina Embeddings v5 + UMAP(10) achieved F1 0.837.

2nd Year: Generative Computer Vision

haritonn/coursework2

Compared diffusion models, GANs, and large multimodal model (LMM) approaches for virtual try-on; evaluated visual quality and inference time on a custom dataset.

Education

Saratov State University

Saratov, Russia

Bachelor | Fundamental Computer Science and Information Technology

September 2023 – June 2027

Teaching & Community

- Teach machine learning in Development Students Clubs (DSC).
- Lecture recordings: NumPy, Pandas ML Club playlist.